A Triphasic Theory for the Swelling and Deformation Behaviors of Articular Cartilage

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Swelling of articular cartilage depends on its fixed charge density and distribution, the stiffness of its collagen-proteoglycan matrix, and the ion concentrations in the interstitium. A theory for a tertiary mixture has been developed, including the two fluid-solid phases (biphasic), and an ion phase, representing cation and anion of a single salt, to describe the deformation and stress fields for cartilage under chemical and/or mechanical loads. This triphasic theory combines the physico-chemical theory for ionic and polyionic (proteoglycan) solutions with the biphasic theory for cartilage. The present model assumes the fixed charge groups to remain unchanged, and that the counter-ions are the cations of a single salt of the bathing solution. The momentum equation for the neutral salt and for the intersitial water are expressed in terms of their chemical potentials whose gradients are the driving forces for their movements. These chemical potentials depend on fluid pressure p, salt concentration c, solid matrix dilatation e and fixed charge density c^F. *For a uni-uni valent salt such as NaCl, they are given by* $\mu^i = \mu_o^i + (RT/M_i)ln[\gamma_\pm^2 c(c + c^F)]$ *and* $\mu^w = \mu_o^w + [p - RT\phi(2c + c^F) + B_w e]/\rho_T^w$, *where R, T,* M_i, $\gamma_\pm$, ϕ, ρ_T^w *and* B_w *are universal gas constant, absolute temperature, molecular weight, mean activity coefficient of salt, osmotic coefficient, true density of water, and a coupling material coefficient, respectively. For infinitesimal strains and material isotropy, the stress-strain relationship for the total mixture stress is* $\sigma = -pI - T_c I + \lambda_s(trE)I + 2\mu_s E$, *where E is the strain tensor and* (λ_s, μ_s) *are the Lamé constants of the elastic solid matrix. The chemical-expansion stress* $(-T_c)$ *derives from the charge-to-charge repulsive forces within the solid matrix. This theory can be applied to both equilibrium and non-equilibrium problems. For equilibrium free swelling problems, the theory yields the well known Donnan equilibrium ion distribution and osmotic pressure equations, along with an analytical expression for the "pre-stress" in the solid matrix. For the confined-compression swelling problem, it predicts that the applied compressive stress is shared by three load support mechanisms: 1) the Donnan osmotic pressure; 2) the chemical-expansion stress; and 3) the solid matrix elastic stress. Numerical calculations have been made, based on a set of equilibrium free-swelling and confined-compression data, to assess the relative contribution of each mechanism to load support. Our results show that all three mechanisms are important in determining the overall compressive stiffness of cartilage.*

Introduction

Articular cartilage changes its dimensions, volume, and weight when the ion concentration in the bathing solution is changed [1–8]. For example, for an unloaded specimen of cartilage soaked in NaCl solutions at constant temperature, the tissue dimensions decrease in an exponential manner with increasing NaCl concentration; this decrease approaches an asymptote at a high concentration, e.g., 2.5 *M* NaCl [6]. Therefore, at the physiological state of 0.15 *M* NaCl, cartilage is in a swollen state, with a swelling pressure resisted by the elastic stress in the collagen-proteoglycan solid matrix. This is often referred to as the collagen prestress or elastic stress in the solid matrix [3, 4, 10, 11]. For a normal cartilage bathed in 0.15 *M* NaCl solution, the swelling pressure has been measured to be around 0.17 MPa [2–4].

In the literature, however, it is less than clear, how swelling pressure, Donnan osmotic pressure or solid matrix stress are related to one another. For example, is swelling pressure the same as Donnan osmotic pressure for cartilage? Is compressive stiffness the same as swelling pressure or Donnan osmotic pressure? Is chemical stress the same as the swelling pressure? We feel that the answers to these questions can be made clear with a theory which includes both the physico-chemical view

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Table 1 Summary of different theoretical models for articular cartilage

Model \ Feature	Constitutive Equation for Solid Matrix	Solid Matrix Deformation and Stress Fields	Interstitial Fluid Flow Field	Ion Diffusion	Effect of FCD on Ion Distribution	Constitutive Equations for chemical Potentials	Interstitial Fluid Pressure Due to Osmotic Effect	Chemical Expansion in Solid Matrix	Remarks
Physico-chemical		✓*		✓*	✓*		✓*		* Homogeneous field only
Elastic and Viscoelastic	✓	✓							
Biphasic/ Poroelastic	✓	✓	✓						
Thermoanalog	✓	✓	✓	✓				✓	
Electro-mechanic	✓	✓	✓	✓	✓		✓*	✓	* Method of calculation not provided
Bicomponent	✓	✓	✓			✓*			* Depends only on matrix dilatation
Triphasic	✓	✓	✓	✓	✓	✓	✓	✓	

[2, 3, 12, 13] and the biphasic or poroelastic view [14–18]. The physico-chemical view of cartilage and other hydrated charged tissue are based on the Donnan (1924) theory for aqueous polyelectrolyte solutions [19] to predict the ion distributions and osmotic pressure [2, 3, 12, 13], while the biphasic or poroelastic view treats the collagen-proteoglycan matrix as a solid with the elastic properties depending on all the underlying physicochemical effects [14–17]. Some attempts have been made to bridge this gap, i.e., the thermoelastic analog theory [5], the electromechanical theory [7, 8] and the bicomponent theory [20]. In our view, all these theories take the ionic effect into consideration in varying but incomplete ways. Thus, a self-contained and self-consistent field theory for the physico-chemical swelling and deformation behavior of cartilage and other hydrated soft tissues is necessary. Table 1 provides a summary of the various theories used to describe cartilage, and those aspects of cartilage physico-chemistry and biomechanics each theory covers.

Origin of Cartilage Swelling. In cartilage, proteoglycan aggregates (PGA) are immobilized and restrained in the collagen meshwork. The PGA's are very large molecules, ranging from 100×10^6 to 200×10^6 daltons [3, 21]. They contain a large number of negatively charged groups (SO_3^- and COO^-) along their glycosaminoglycan (GAG) chains. The density of these charges is called the fixed charge density (*FCD*). For example, for normal and degenerate femoral head cartilage, the *FCD* ranges from 0.04 to 0.18 mEq/(g wet tissue) at the physiological pH level [2, 3]. Each negative charge requires a counter-ion to be nearby to maintain electro-neutrality. When bathed in a very dilute ionic solution, such as a 0.0015 *M* NaCl solution, the concentration (c^F) of the univalent counter-ion (e.g., Na^+) in the tissue is essentially equal to the *FCD* [9]. The total ion concentration inside the tissue is always greater than the ion concentration in the external bathing solution. This imbalance of ions gives rise to a pressure in the interstitial fluid which is higher than the ambient pressure in the external bath. This is known as the Donnan (ionic) osmotic pressure. This osmotic pressure is one of the causes of cartilage swelling. 渗透压

In the extracellular matrix (ECM), it is estimated that the PGA's may be restrained to 1/5 of their volume in free solution [21, 22]. Thus, the fixed charge groups along the GAG chains are very close to each other, causing charge-to-charge repulsive forces to be exerted against each other. These forces may also cause the ECM to swell. The charge-to-charge electrostatic forces are modulated by the counter- and co-ions swarming around the PG molecules in solution. With increasing ion concentrations, the equivalent Debye length between the ion cloud and the fixed charges is decreased. This results in charge shielding which decreases the net charge-to-charge repulsive force [23]. Decrease of the size of the PG molecule in solution with charge shielding has been measured [24]. Changes in the dimensions of the tissue specimen, and isometric stress, due to NaCl concentration changes in the external bathing solution have also been measured [1, 5–8, 25, 26]. Thus, one of the objectives of the present investigation is the determination of how these two coupled effects, the Donnan osmotic pressure effect (excess ions in the interstitial water) and the chemical expansion effect (electrostatic repulsive forces exerted on the PG-collagen network) contribute to the swelling behavior of cartilage and to its mechanical stiffness in compression. 电荷斥力

Chemical Potentials and Driving Forces. While physico-chemical theories for osmotic pressures in polyelectrolytic solutions, Fig. 1(*a*), are adequate and well accepted [19, 27, 28], a field theory for charged fiber-reinforced fluid-solid gels, Fig. 1(*b*), one in which physicochemical laws have been incorporated into a continuum mixture formulation of constitutive and field equations, appears to be lacking [5, 7, 8, 20]. To develop this mechano-electrochemical field theory, we make use of the fact that the driving forces for the movements of water and neutral salt in these charged gels are gradients of their chemical potentials [27]. For a specific charged gel such as the ECM of cartilage, specific constitutive assumptions for

Nomenclature

a_o = coefficient in chemical-expansion stress
A^α = Helmholtz free energy per unit mass of α-component
$\mathbf{b}^\alpha$ = body force on unit mass of α-component
B_i = coupling coefficients in chemical potentials
B_w = coupling coefficients in chemical potentials
c = neutral salt concentration in mole per unit tissue water volume
c^* = salt concentration in bath, in mole per unit volume
$\mathbf{C}$ = right Cauchy-Green deformation tensor of solid matrix
c^F = fixed charge density (Eq/unit tissue water volume)
$\tilde{c}^F$ = fixed charge density (Eq/unit tissue volume)
$c^\pm$ = charge density for cations (+), anions (−) (Eq/tissue water volume)
$\tilde{c}^\pm$ = charge density for cations (+), anions (−) (Eq/tissue volume)
e_{zz} = axial strain from hypertonic reference configuration
e^α = internal energy per unit mass of α-component
$\mathbf{E}$ = infinitesimal deformation tensor of solid matrix
$\mathbf{E}_e$ = electric field
$\mathbf{F}$ = deformation gradient of solid matrix
F_c = Faraday constant
H_A = confined compression modulus at given c^*
$\mathbf{K}^\alpha$ = chemical potential tensor of α-component
$f_{\alpha\beta}$ = drag coefficients between α and β components
$K_{\alpha\beta}$ = drag coefficients in triphasic equations
$\mathbf{L}^\alpha$ = velocity gradient of α-component
$M^\pm$ = atomic weight of Na and Cl
M_i = molecular weight of NaCl
p = fluid pressure in tissue
P_s = total swelling pressure
P_∞ = osmotic pressure from colligative effect of PG particles in tissue

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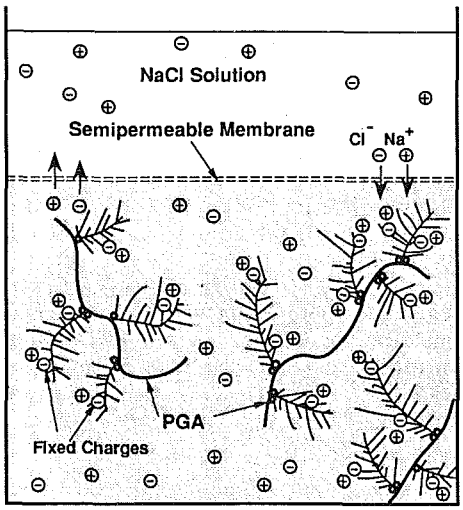

Fig. 1(*a*) Schematic diagram of a proteoglycan solution equilibrated with NaCl solution, separated by a semi-permeable membrane

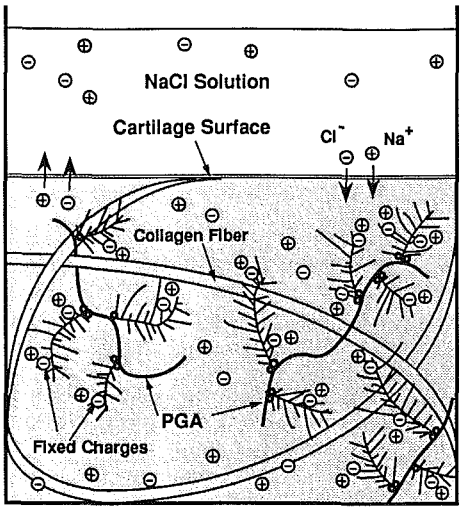

Fig. 1(*b*) Schematic diagram of articular cartilage equilibrated with NaCl solution

its chemical potential and strain energy potential are necessary. For cartilage, attempts have been made using chemical potentials to derive such a theory [29]. These investigators proposed a "state variable" model in which they regard as "the only relevant chemical potential [to be] the potential of the reference medium."[1] However, no chemical potentials for either the fluid or the ions in the tissue were introduced.[2] In contrast, in the present study, we propose a three phase model for cartilage which is composed of (1) the interstitial fluid phase; (2) the collagen-proteoglycan solid matrix phase; and (3) a second fluid phase—the ionic phase. The ionic phase is further assumed to be consisting of two ionic species—those of a single salt (such as NaCl) and that all the counter-ions are the cations of the salt. In this triphasic theory, the stresses in the solid matrix and the chemical potentials for the interstitial fluid and ions are related to the Helmholtz energy functions in accordance with the laws of energy balance and entropy.

Modern continuum mixture theory started with C. Truesdell [30–32]. Other important contributors to the theory of mixture may be found in [32]. Many specialized theories have since been developed, in particular, the theory for a mixture of one incompressible solid phase and N-1 incompressible fluid phases was developed in 1980 by Bowen [33]. This provides the basic framework for our development of our triphasic theory. In this theory, the Helmholtz energy functions for the phases depend on the deformation of the solid phase, temperature, and the densities of the N-1 fluid phases. In the present study, this theoretical framework is specialized to our study of articular cartilage which contains a elastic solid matrix, a fluid phase (water) and a third phase for the ionic species of the salt. However, to include the fixed charge groups of the solid matrix of cartilage, we specifically introduce into the Helmholtz energy functions a dependence on c^F. In addition, for the ion phase and the fluid phase, we incorporate the traditional chemical potentials for electrolyte and polyelectrolyte solutions into the theory, thus providing a theoretical bridge between the continuum mixture theory applied to soft tissues [14, 16] and the physico-chemical theory [2, 3, 12, 13, 19, 27, 28]. Hence, it will be useful that we briefly summarize the portion

[1]According to [29], "fluid within cartilage is hardly dilute enough to use the simple relation between concentration and chemical potential," and for the case of two monovalent solutes with concentrations c^+ and c^-, the chemical potential for the reference medium was given by $\mu = RT\ln(c^+ + c^-)$.

[2]In our view, it is not possible to study either the equilibrium or the kinetic swelling behaviors of cartilage using a single chemical potential for the whole tissue.

Nomenclature (cont.)

$\mathbf{q}^\alpha$ = heat flux vector in α-component
r^α = internal heat supply to α-component
R = universal gas constant
T = absolute temperature
$-T_c\mathbf{I}$ = chemical-expansion stress in solid phase
$\mathbf{v}^\alpha$ = velocity of α-component
x_α = molar fraction
$\gamma_\pm$ = mean activity coefficient of ions
δ^α = total energy supply to α-component
ϵ = compressive axial strain $(= -e_{zz})$
ϵ' = compressive strain measured from unloaded configuration at c^*
ϵ^α = internal energy supply to α-component
η^α = entropy per unit mass of α-component
λ_s, μ_s = Lamé constants of solid matrix
μ^α = chemical potential per unit mass of α-component $(\alpha = w, +, -)$
μ^i = chemical potential per unit mass of the ion phase (cations and anions)
ν_α = chemical potential per mole of α-component $(\alpha = w, i)$
ϕ^α = volume fraction of α-component
π^α = momentum supply to α-component
ρ^α = apparent mass density of α-component (per unit tissue volume)
ρ^F = apparent mass density of cations needed to neutralize *FCD*
ρ^i = apparent mass density of neutral salt $= \rho^+ + \rho^- - \rho^F$
ρ_T^α = true density of α-component (per unit component volume)
σ^α = stress tensor of α-component
σ = stress tensor of mixture (tissue) $= \sigma^s + \sigma^w + \sigma^+ + \sigma^-$
σ^f = fluid stress tensor (water and ions) $= \sigma^w + \sigma^+ + \sigma^-$
ϕ = osmotic coefficient

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of the Donnan equilibrium osmotic theory which relates to our proposed triphasic theory for articular cartilage, and other charged hydrated soft tissues.

The Donnan Theory of Ionic Equilibria and Osmotic Pressure. When two NaCl solutions of different concentrations are separated by a membrane which is permeable to both Na^+ and Cl^- ions, as well as the common solvent, the equilibrium concentrations of NaCl are equal on both sides of the membrane. However, if on one side (defined as the internal solution) the NaCl solution also contains polyelectrolytes (e.g., PGA's) which are too large to move to the other solution (defined as the external solution) through the membrane, Fig. 1*a*, then the concentration of Na^+ and Cl^- ions of the two sides will be different. The concentration of Na^+ required to balance the negative charges on the GAG chains of the PG molecules in the internal solution required by the electroneutrality condition is denoted by c^F. So long as *FCD* in the internal solution does not change, c^F is a constant, i.e., independent of the NaCl concentration c^* in the external solution. In addition to these Na^+ in the internal solution, there are also Cl^- and an equal amount of Na^+ whose concentration varies with c^*. If c denotes the concentration of this varying portion of Na^+, then the total Na^+ concentration (c_{Na^+}) is $(c+c^F)$ and the total Cl^- concentration (c_{Cl^-}) is c. In the external solution, $c^* = c^*_{Na^+} = c^*_{Cl^-}$.

Let the chemical potential per mole of the NaCl in the external solution be denoted by ν^*_{NaCl} and that in the internal solution by ν_{NaCl}. These chemical potentials are given by [27]:

$$\nu^*_{NaCl} = \nu^*_o(T) + RT\ln(\gamma^*_+ c^*_{Na} + \gamma^*_- c^*_{Cl^-}) = \nu^*_o + RT\ln(\gamma^{*2}_\pm c^{*2}), \quad (1)$$

$$\nu_{NaCl} = \nu_o(T) + RT\ln(\gamma_+ c_{Na} + \gamma_- c_{Cl^-}) = \nu_o + RT\ln[\gamma_\pm^2 c(c+c^F)], \quad (2)$$

where R is the universal gas constant, T the absolute temperature, $\gamma_\pm$ and $\gamma^*_\pm$ the mean activity coefficients with $\gamma_\pm^2 = \gamma_+\gamma_-$ and $\gamma^{*2}_\pm = \gamma^*_+\gamma^*_-$, ν_o and ν^*_o the concentration-independent potential and $\nu_o = \nu^*_o$. When the two solutions are at equilibrium, their chemical potential must be equal, i.e., $\nu^*_{NaCl} = \nu_{NaCl}$. This leads to the following equation for equilibrium ion distribution between the external and internal solutions:

$$\frac{c^{*2}}{c(c+c^F)} = \left(\frac{\gamma_\pm}{\gamma^*_\pm}\right)^2. \quad (3)$$

For the ideal solutions $\gamma_\pm = \gamma^*_\pm = 1$, the above equation reduces to the classical Donnan ion distribution equation. The activity coefficients $\gamma_\pm$ and $\gamma^*_\pm$ generally depend on c (or c^*) and account for the interactions among the ions and the macromolecules in real solutions. Maroudas, who measured c and c^F for cartilage slices equilibrated in $c^* = 0.15\ M$ NaCl, showed that $\gamma_\pm/\gamma^*_\pm$ is close to one for femoral head and condyle cartilage [2, 3].

From Eq. (3), we see that when the external solution ion concentration approaches zero ($c^* \to 0$), the concentration c of NaCl in the internal solution also approaches zero. There are however the counter-ions at concentration c^F which equals the *FCD* as required by electroneutrality. Using c^F in the van't Hoff equation for dilute solutions, we obtain $p = RTc^F$ for the Donnan (ionic) osmotic pressure due to the ions associated with the *FCD*. Thus, if the external solution is a very dilute salt solution, we have a maximal osmotic pressure effect due to the *FCD*. If the external solution is hypertonic, i.e., c^*/c^F is large, then Eq. (3) yields:

级数

$$c \simeq c^* - \frac{c^F}{2} + \frac{(c^F)^2}{8c^*}. \quad (4)$$

We see that the concentration of ions in the internal solution $(2c+c^F)$ exceeds that in the external solution $(2c^*)$ by $(c^F)^2/4c^*$. Thus, in hypertonic solutions, the osmotic pressure may be approximated by the relationship

$$p = RT\frac{(c^F)^2}{4c^*}. \quad (5)$$

This pressure approaches zero at the rate of $1/c^*$ as c^* becomes large. These results were developed by Donnan in 1924 [19, also see 28], and used extensively by Maroudas [2, 3] and Urban and co-workers [12, 13] for articular cartilage and intervertebral disk, respectively.

In the present study, we will develop a triphasic theory for cartilage which incorporates the equilibrium Donnan ion distribution and osmotic pressure theory for polyelectrolytic solutions. This triphasic theory will also subsume the biphasic theory for the deformational and flow behaviors for the tissue.

A Triphasic Model

Consider a mixture of three phases: an incompressible solid phase (the collagen-proteoglycan ECM), an incompressible fluid phase (interstitial water) and an ion phase of two species (cations and anions) of a single salt (e.g., NaCl). Our model assumes that the counter-ions associated with the *FCD* are the same kind of cations of the salt (e.g., Na^+). Let ϕ^α denote the volume fraction of each component[3] (i.e., $\phi^\alpha = dV^\alpha/dV$), we have

$$\sum_{s,w,+,-} \phi^\alpha = 1 \quad (6)$$

where $\phi^s = dV^s/dV$ is the solidity, $\phi^w = dV^w/dV$ the porosity of the tissue, and ϕ^+ and ϕ^- are very small compared with ϕ^s and ϕ^w.

Continuity Equations and Electroneutrality Condition. When no chemical reaction takes place, the mass balance for each component is

$$\frac{\partial \rho^\alpha}{\partial t} + \mathrm{div}(\rho^\alpha \mathbf{v}^\alpha) = 0, \quad (7)$$

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where $\alpha = s, w, +, -$, denote the solid matrix, interstitial water, cations Na^+ and anions Cl^- respectively, and ρ^α is the apparent density (mass of α-component per unit dV), related to their true densities (mass of α-component per unit volume of the α-component) by $\rho^\alpha = \phi^\alpha \rho^\alpha_T$. From Eqs. (6) and (7), and the assumption of incompressibility, one obtains the equation of continuity for the mixture:

$$\mathrm{div}(\phi^s \mathbf{v}^s + \phi^w \mathbf{v}^w + \phi^+ \mathbf{v}^+ + \phi^- \mathbf{v}^-) = 0 \quad (8)$$

The electoneutrality condition is expressed by the equation:

$$\tilde{c}^+ = \tilde{c}^- + \tilde{c}^F \quad (9)$$

mol/L

where $\tilde{c}^+$, $\tilde{c}^-$ and $\tilde{c}^F$ are the charge density (Eq/tissue volume) for cation, anion and the fixed charges, respectively. This equation is to be true for all time. With M^+ and M^- denoting the atomic weights for Na^+ and Cl^-, respectively, the charges per unit tissue volume and the mass densities are related by:

原子量

g/mol

$$\rho^+ = M^+\tilde{c}^+ \text{ and } \rho^- = M^-\tilde{c}^- \quad (10)$$

g/L

Thus, from the equations of continuity for ρ^+ and ρ^-, we have:

$$\frac{\partial \tilde{c}^+}{\partial t} + \mathrm{div}(\tilde{c}^+ \mathbf{v}^+) = 0 \text{ and } \frac{\partial \tilde{c}^-}{\partial t} + \mathrm{div}(\tilde{c}^- \mathbf{v}^-) = 0. \quad (11)$$

Since in our model, we assume the fixed charge groups to remain unchanged, $\tilde{c}^F$ must be conserved at all time during deformation of the tissue. Thus,

$$\frac{\partial \tilde{c}^F}{\partial t} + \mathrm{div}(\tilde{c}^F \mathbf{v}^s) = 0. \quad (12)$$

[3]Even though ions are miscible with the interstital water, it is simple to regard them as immiscible with very small ϕ^α.

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Combining Eqs. (9), (11), and (12), we obtain

$$\operatorname{div} \mathbf{I}_e = 0 \tag{13}$$

where $\mathbf{I}_e = F_c[\tilde{c}^+(\mathbf{v}^+ - \mathbf{v}^s) - \tilde{c}^-(\mathbf{v}^- - \mathbf{v}^s)]$ is the electric current density and F_c is the Faraday constant. When the current density is zero, or neglected, the velocities of the anions and the cations are related by:

$$\tilde{c}^+(\mathbf{v}^+ - \mathbf{v}^s) = \tilde{c}^-(\mathbf{v}^- - \mathbf{v}^s). \tag{14}$$

Momentum, Energy and Entropy Laws. The balance of linear momentum for each component leads to the following equations of motion:

$$\rho^\alpha \frac{D^\alpha \mathbf{v}^\alpha}{Dt} = \operatorname{div}\boldsymbol{\sigma}^\alpha + \rho^\alpha \mathbf{b}^\alpha + \boldsymbol{\pi}^\alpha, \tag{15}$$

where D^α/Dt is the material derivative following the α-component, defined as $\partial/\partial t + \mathbf{v}^\alpha \cdot \nabla$, $\boldsymbol{\sigma}^\alpha$ is the apparent stress tensor, $\mathbf{b}^\alpha$ is the external body force per unit mass, and $\boldsymbol{\pi}^\alpha$ is the momentum supply to the α-component from the other components. When the effect of electric field on the motion of each individual ionic species is of interest, the body force $\rho^\alpha \mathbf{b}^\alpha = z_\alpha F_c \tilde{c}^\alpha \mathbf{E}_e$ with $z_{Na^+} = +1$, $z_{Cl^-} = z_s = -1$, $z_w = 0$ and $\mathbf{E}_e$ is the electrical field. Otherwise, the body force may be taken to be zero. By the conservation of linear momentum for the mixture, we must have

$$\boldsymbol{\pi}^s + \boldsymbol{\pi}^w + \boldsymbol{\pi}^+ + \boldsymbol{\pi}^- = \mathbf{0}. \tag{16}$$

Assuming no local supply of moment of momentum, then, from the balance of moment of momentum, we can conclude that the stress tensors are symmetric.

The balance of energy for each component yields the following equations for the internal energy density e^α:

$$\rho^\alpha \frac{D^\alpha e^\alpha}{Dt} = tr(\boldsymbol{\sigma}^\alpha \cdot \mathbf{L}^\alpha) - \operatorname{div}\mathbf{q}^\alpha + \rho^\alpha r^\alpha + \epsilon^\alpha, \tag{17}$$

where $\mathbf{L}^\alpha$ is the velocity gradient, $\mathbf{q}^\alpha$ the heat flux vector, r^α the internal heat supply, and ϵ^α is the part of the energy supply to the α-component from the other components which contributes to the increase of the internal energy. The total energy supply to each of the α-component is $\delta^\alpha = \epsilon^\alpha + \boldsymbol{\pi}^\alpha \cdot \mathbf{v}^\alpha$, and this must satisfy the energy conservation equation for the total mixture:

$$\delta^s + \delta^w + \delta^+ + \delta^- = 0. \tag{18}$$

The second law of thermodynamics for the mixture is expressed by:

$$\sum_{\alpha=s,w,+,-} \left[\rho^\alpha \frac{D^\alpha \eta^\alpha}{Dt} + \operatorname{div}\left(\frac{\mathbf{q}^\alpha}{T}\right) - \rho^\alpha \frac{r^\alpha}{T} \right] \geq 0, \tag{19}$$

where η^α is the entropy density of α-component, and T is the absolute temperature. By the energy Eq. (17), this inequality can be written:

$$\sum_{\alpha=s,w,+,-} \left[-\rho^\alpha \left(\frac{D^\alpha A^\alpha}{Dt} + \eta^\alpha \frac{D^\alpha T}{Dt} \right) + tr(\boldsymbol{\sigma}^\alpha \cdot \mathbf{L}^\alpha) - \frac{1}{T}\mathbf{q}^\alpha \cdot \nabla T - \mathbf{v}^\alpha \cdot \boldsymbol{\pi}^\alpha \right] \geq 0, \tag{20}$$

where the Helmholtz energy density A^α is defined by

$$A^\alpha = e^\alpha - T\eta^\alpha. \tag{21}$$

For a triphasic medium with an elastic matrix containing fixed charges undergoing finite deformation, we make the following constitutive assumption on A^α for cartilage and other hydrated soft tissues:

$$A^\alpha = A^\alpha(T, \mathbf{C}, \rho^w, \rho^+, \rho^-, c^F), \tag{22}$$

where $\mathbf{C}$ is the right Cauchy-Green deformation tensor of the solid matrix. Again, we assume that the fixed charge groups per unit mass of the solid phase remain a constant at all time, so that c^F is solely determined by its value c_o^F at the reference configuration for deformation and the volume ratio $f \equiv dV_o/dV(t)$, which is a function of $\det\mathbf{C}$,[4]

$$c^F = c_o^F f(\det\mathbf{C}). \tag{23}$$

The variables $\mathbf{L}^\alpha$ and $\mathbf{v}^\alpha - \mathbf{v}^\beta$ ($\alpha, \beta = s, w, +, -$), which will appear in the entropy inequality when Eq. (22) is substituted in (20), are subjected to two constraints: (i) the incompressibility assumption Eq. (8), which can be written as:

$$\sum_{\alpha=s,w,+,-} \phi^\alpha tr(\mathbf{L}^\alpha) + \sum_{\beta=w,+,-} (\mathbf{v}^\beta - \mathbf{v}^s) \cdot \nabla \phi^\beta = 0 \tag{24}$$

and (ii) the three independent identity relations between the six relative velocities, e.g., $(\mathbf{v}^s - \mathbf{v}^w) + (\mathbf{v}^w - \mathbf{v}^+) + (\mathbf{v}^+ - \mathbf{v}^-) + (\mathbf{v}^- - \mathbf{v}^s) = \mathbf{0}$, etc.

These two types of constraints must be incorporated into the entropy inequality. The first condition imposes a constraint on the material responses and therefore must be taken into account in the inequality. The second condition allows one to treat all the relative velocities as independent variables in the entropy inequality.

With these constraints the entropy inequality (20) becomes:[5]

$$\begin{aligned}
&\sum_{s,w,+,-} \left[-\rho^\alpha \left(\frac{\partial A^\alpha}{\partial T} + \eta^\alpha \right) \frac{D^\alpha T}{Dt} - \frac{1}{T}\mathbf{q}^\alpha \cdot \nabla T \right] \\
&+ tr\left\{ \left[\boldsymbol{\sigma}^s + \phi^s \lambda \mathbf{I} - 2\mathbf{F}\left(\rho^s \frac{\partial A^s}{\partial \mathbf{C}} + \rho^w \frac{\partial A^w}{\partial \mathbf{C}} + \rho^+ \frac{\partial A^+}{\partial \mathbf{C}} + \rho^- \frac{\partial A^-}{\partial \mathbf{C}} \right) \mathbf{F}^T \right.\right. \\
&\left.\left. - 2c_o^F(\det\mathbf{C}) f' \left(\rho^s \frac{\partial A^s}{\partial c^F} + \rho^w \frac{\partial A^w}{\partial c^F} + \rho^+ \frac{\partial A^+}{\partial c^F} + \rho^- \frac{\partial A^-}{\partial c^F} \right) \mathbf{I} \right) \mathbf{L}^s \right\} \\
&+ tr\left\{ \left[\boldsymbol{\sigma}^w + \phi^w \lambda \mathbf{I} + \rho^w \left(\rho^s \frac{\partial A^s}{\partial \rho^w} + \rho^w \frac{\partial A^w}{\partial \rho^w} + \rho^+ \frac{\partial A^+}{\partial \rho^w} + \rho^- \frac{\partial A^-}{\partial \rho^w} \right) \mathbf{I} \right] \mathbf{L}^w \right\} \\
&+ tr\left\{ \left[\boldsymbol{\sigma}^+ + \phi^+ \lambda \mathbf{I} + \rho^+ \left(\rho^s \frac{\partial A^s}{\partial \rho^+} + \rho^w \frac{\partial A^w}{\partial \rho^+} + \rho^+ \frac{\partial A^+}{\partial \rho^+} + \rho^- \frac{\partial A^-}{\partial \rho^+} \right) \mathbf{I} \right] \mathbf{L}^+ \right\} \\
&+ tr\left\{ \left[\boldsymbol{\sigma}^- + \phi^- \lambda \mathbf{I} + \rho^- \left(\rho^s \frac{\partial A^s}{\partial \rho^-} + \rho^w \frac{\partial A^w}{\partial \rho^-} + \rho^+ \frac{\partial A^+}{\partial \rho^-} + \rho^- \frac{\partial A^-}{\partial \rho^-} \right) \mathbf{I} \right] \mathbf{L}^- \right\}
\end{aligned}$$

[4]For example, if c^F is concentration per unit tissue volume, then $f(\det\mathbf{C}) = [\det\mathbf{C}]^{-1/2}$. A different f results if c^F is concentration per unit volume of tissue water.

[5]$[\nabla\mathbf{P}]\mathbf{Q}$ denotes the vector given by $\partial_i P_{mn} Q_{mn} \mathbf{e}_i$ for any two tensors $\mathbf{P}$ and $\mathbf{Q}$.

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$$+\rho^s\frac{\partial A^s}{\partial\rho^w}(\mathbf{v}^w-\mathbf{v}^s)\cdot\nabla\rho^w+\rho^s\frac{\partial A^s}{\partial\rho^+}(\mathbf{v}^+-\mathbf{v}^s)\cdot\nabla\rho^+ +\rho^s\frac{\partial A^s}{\partial\rho^-}(\mathbf{v}^- -\mathbf{v}^s)\cdot\nabla\rho^-$$

$$+\rho^w\frac{\partial A^w}{\partial\rho^+}(\mathbf{v}^+-\mathbf{v}^w)\cdot\nabla\rho^+ +\rho^w\frac{\partial A^w}{\partial\rho^-}(\mathbf{v}^- -\mathbf{v}^w)\cdot\nabla\rho^- +\rho^w(\mathbf{v}^s-\mathbf{v}^w)\cdot[\nabla\mathbf{C}]\frac{\partial A^w}{\partial\mathbf{C}}$$

$$+\rho^+\frac{\partial A^+}{\partial\rho^w}(\mathbf{v}^w-\mathbf{v}^+)\cdot\nabla\rho^w+\rho^+\frac{\partial A^+}{\partial\rho^-}(\mathbf{v}^- -\mathbf{v}^+)\cdot\nabla\rho^- +\rho^+(\mathbf{v}^s-\mathbf{v}^+)\cdot[\nabla\mathbf{C}]\frac{\partial A^+}{\partial\mathbf{C}}$$

$$+\rho^-\frac{\partial A^-}{\partial\rho^w}(\mathbf{v}^w-\mathbf{v}^-)\cdot\nabla\rho^w+\rho^-\frac{\partial A^-}{\partial\rho^+}(\mathbf{v}^+-\mathbf{v}^-)\cdot\nabla\rho^+ +\rho^-(\mathbf{v}^s-\mathbf{v}^-)\cdot[\nabla\mathbf{C}]\frac{\partial A^-}{\partial\mathbf{C}}\cdot$$

$$+c_o^F(\det\mathbf{C})f'\left[\rho^w\frac{\partial A^w}{\partial c^F}(\mathbf{v}^s-\mathbf{v}^w)+\rho^+\frac{\partial A^-}{\partial c^F}(\mathbf{v}^s-\mathbf{v}^+)+\rho^-\frac{\partial A^-}{\partial c^F}(\mathbf{v}^s-\mathbf{v}^-)\right]\cdot[\nabla\mathbf{C}]\mathbf{C}^{-1}$$

$$-(\pi^s-\lambda\nabla\phi^s)\cdot\mathbf{v}^s-(\pi^w-\lambda\nabla\phi^w)\cdot\mathbf{v}^w-(\pi^+-\lambda\nabla\phi^+)\cdot\mathbf{v}^+-(\pi^- -\lambda\nabla\phi^-)\cdot\mathbf{v}^-$$

$$+\eta_1\cdot[(\mathbf{v}^s-\mathbf{v}^w)+(\mathbf{v}^w-\mathbf{v}^+)+(\mathbf{v}^+-\mathbf{v}^-)+(\mathbf{v}^- -\mathbf{v}^s)]$$

$$+\eta_2\cdot[(\mathbf{v}^s-\mathbf{v}^+)+(\mathbf{v}^+-\mathbf{v}^w)+(\mathbf{v}^w-\mathbf{v}^s)]$$

$$+\eta_3\cdot[(\mathbf{v}^w-\mathbf{v}^-)+(\mathbf{v}^- -\mathbf{v}^+)+(\mathbf{v}^+-\mathbf{v}^w)]\geq 0, \tag{25}$$

where $\mathbf{F}$ is the deformation gradient, and λ and η's are Lagrangian multipliers. This inequality must be satisfied for all thermodynamic processes, thus, we obtain the constitutive relations for the stresses and momentum supplies:

$$\sigma^s=-\phi^s\lambda\mathbf{I}+2c_o^F(\det\mathbf{C})f'\left(\rho^s\frac{\partial A^s}{\partial c^F}+\rho^w\frac{\partial A^w}{\partial c^F}+\rho^+\frac{\partial A^+}{\partial c^F}+\rho^-\frac{\partial A^-}{\partial c^F}\right)\mathbf{I}+2\mathbf{F}\left(\rho^s\frac{\partial A^s}{\partial\mathbf{C}}+\rho^w\frac{\partial A^w}{\partial\mathbf{C}}+\rho^+\frac{\partial A^+}{\partial\mathbf{C}}+\rho^-\frac{\partial A^-}{\partial\mathbf{C}}\right)\mathbf{F}^T, \tag{26a}$$

$$\sigma^w=-\phi^w\lambda\mathbf{I}-\rho^w\left(\rho^s\frac{\partial A^s}{\partial\rho^w}+\rho^w\frac{\partial A^w}{\partial\rho^w}+\rho^+\frac{\partial A^+}{\partial\rho^w}+\rho^-\frac{\partial A^-}{\partial\rho^w}\right)\mathbf{I}, \tag{26b}$$

$$\sigma^+=-\phi^+\lambda\mathbf{I}-\rho^+\left(\rho^s\frac{\partial A^s}{\partial\rho^+}+\rho^w\frac{\partial A^w}{\partial\rho^+}+\rho^+\frac{\partial A^+}{\partial\rho^+}+\rho^-\frac{\partial A^-}{\partial\rho^+}\right)\mathbf{I}, \tag{26c}$$

$$\sigma^-=-\phi^-\lambda\mathbf{I}-\rho^-\left(\rho^s\frac{\partial A^s}{\partial\rho^-}+\rho^w\frac{\partial A^w}{\partial\rho^-}+\rho^+\frac{\partial A^+}{\partial\rho^-}+\rho^-\frac{\partial A^-}{\partial\rho^-}\right)\mathbf{I}, \tag{26d}$$

$$\pi^s=\lambda\nabla\phi^s-\rho^s\frac{\partial A^s}{\partial\rho^w}\nabla\rho^w+\rho^w[\nabla\mathbf{C}]\frac{\partial A^w}{\partial\mathbf{C}}-\rho^s\frac{\partial A^s}{\partial\rho^+}\nabla\rho^+ -\rho^s\frac{\partial A^s}{\partial\rho^-}\nabla\rho^- +\rho^+[\nabla\mathbf{C}]\frac{\partial A^+}{\partial\mathbf{C}}+\rho^-[\nabla\mathbf{C}]\frac{\partial A^-}{\partial\mathbf{C}}$$
$$+c_o^F(\det\mathbf{C})f'\left(\rho^w\frac{\partial A^w}{\partial c^F}+\rho^+\frac{\partial A^+}{\partial c^F}+\rho^-\frac{\partial A^-}{\partial c^F}\right)[\nabla\mathbf{C}]\mathbf{C}^{-1}+\mathbf{f}_{sw}(\mathbf{v}^w-\mathbf{v}^s)+\mathbf{f}_{+s}(\mathbf{v}^+-\mathbf{v}^s)+\mathbf{f}_{-s}(\mathbf{v}^- -\mathbf{v}^s), \tag{27a}$$

$$\pi^w=\lambda\nabla\phi^w+\rho^s\frac{\partial A^s}{\partial\rho^w}\nabla\rho^w-\rho^w[\nabla\mathbf{C}]\frac{\partial A^w}{\partial\mathbf{C}}$$
$$-c_o^F(\det\mathbf{C})f'\rho^w\frac{\partial A^w}{\partial c^F}[\nabla\mathbf{C}]\mathbf{C}^{-1}+\rho^+\frac{\partial A^+}{\partial\rho^w}\nabla\rho^w+\rho^-\frac{\partial A^-}{\partial\rho^w}\nabla\rho^w -\rho^w\frac{\partial A^w}{\partial\rho^+}\nabla\rho^+-\rho^w\frac{\partial A^w}{\partial\rho^-}\nabla\rho^- +\mathbf{f}_{sw}(\mathbf{v}^s-\mathbf{v}^w)$$
$$+\mathbf{f}_{w+}(\mathbf{v}^+-\mathbf{v}^w)+\mathbf{f}_{w-}(\mathbf{v}^- -\mathbf{v}^w), \tag{27b}$$

$$\pi^\alpha=\lambda\nabla\phi^\alpha+\rho^s\frac{\partial A^s}{\partial\rho^\alpha}\nabla\rho^\alpha-\rho^\alpha[\nabla\mathbf{C}]\frac{\partial A^\alpha}{\partial\mathbf{C}}$$
$$-c_o^F(\det\mathbf{C})f'\rho^\alpha\frac{\partial A^\alpha}{\partial c^F}[\nabla\mathbf{C}]\mathbf{C}^{-1}-\rho^\alpha\frac{\partial A^\alpha}{\partial\rho^w}\nabla\rho^w+\rho^w\frac{\partial A^w}{\partial\rho^\alpha}\nabla\rho^\alpha +\rho^\beta\frac{\partial A^\beta}{\partial\rho^\alpha}\nabla\rho^\alpha-\rho^\alpha\frac{\partial A^\alpha}{\partial\rho^\beta}\nabla\rho^\beta+\mathbf{f}_{\alpha s}(\mathbf{v}^s-\mathbf{v}^\alpha)$$
$$+\mathbf{f}_{w\alpha}(\mathbf{v}^w-\mathbf{v}^\alpha)+\mathbf{f}_{\alpha\beta}(\mathbf{v}^\beta-\mathbf{v}^\alpha), \tag{27c,d}$$

where f' denotes differentiation with respect to its argument.[6] In Eq. (27*c*), $\alpha=+$, $\beta=-$ and in Eq. (27*d*), $\alpha=-$, $\beta=+$. Also, $\mathbf{f}_{\alpha\beta}$ $(\alpha,\beta=s,w,+,-)$ are positive definite tensors whose elements are the frictional drag coefficients between the α- and β-component. Following the classical mixture theories, we assume $\mathbf{f}_{\alpha\beta}=\mathbf{f}_{\beta\alpha}$ and for isotropic mixtures, $\mathbf{f}_{\alpha\beta}=\mathbf{f}_{\alpha\beta}\mathbf{I}=\mathbf{f}_{\beta\alpha}\mathbf{I}$.[7]

Equations of Motion in Terms of Chemical Potentials. The chemical potential tensor of each component is defined as follows

$$\mathbf{K}^s=A^s\mathbf{I}-\sigma^s/\rho^s,\ \mathbf{K}^w=\mu^w\mathbf{I}=A^w\mathbf{I}-\sigma^w/\rho^w,\ \mathbf{K}^\pm=\mu^\pm\mathbf{I}=A^\pm\mathbf{I}-\sigma^\pm/\rho^\pm. \tag{28}$$

From Eqs. (26), we see that σ^w and $\sigma^\pm$ are isotropic tensors so that the chemical potentials $\mathbf{K}^w$, $\mathbf{K}^+$ and $\mathbf{K}^-$ reduce to scalar potentials μ^w, μ^+ and μ^-. In terms of these quantities, the quasi-static momentum equations (15) become:

$$-\rho^s\mathrm{div}\mathbf{K}^s+\frac{1}{2}\rho^s\mathbf{K}^s([\nabla\mathbf{C}]\mathbf{C}^{-1})-\frac{1}{2}\rho^s[\nabla\mathbf{C}](\mathbf{F}^{-1}\mathbf{K}^s\mathbf{F}^{-T})+f_{sw}(\mathbf{v}^w-\mathbf{v}^s)+f_{+s}(\mathbf{v}^+-\mathbf{v}^s)+f_{-s}(\mathbf{v}^- -\mathbf{v}^s)-F_c\tilde{c}^F\mathbf{E}_e=\mathbf{0}, \tag{29}$$

$$-\rho^w\nabla\mu^w+f_{sw}(\mathbf{v}^s-\mathbf{v}^w)+f_{w+}(\mathbf{v}^+-\mathbf{v}^w)+f_{w-}(\mathbf{v}^- -\mathbf{v}^w)=\mathbf{0}, \tag{30}$$

$$-\rho^+\nabla\mu^++f_{+s}(\mathbf{v}^s-\mathbf{v}^+)+f_{w+}(\mathbf{v}^w-\mathbf{v}^+)+f_{+-}(\mathbf{v}^- -\mathbf{v}^+)+F_c(\rho^+/M^+)\mathbf{E}_e=\mathbf{0}, \tag{31}$$

$$-\rho^-\nabla\mu^-+f_{-s}(\mathbf{v}^s-\mathbf{v}^-)+f_{w-}(\mathbf{v}^w-\mathbf{v}^-)+f_{+-}(\mathbf{v}^+-\mathbf{v}^-)-F_c(\rho^-/M^-)\mathbf{E}_e=\mathbf{0}. \tag{32}$$

Triphasic Momentum Equations in Terms of Chemical Potentials. In the absence of electrical current or if the effect of current is neglected, the speed of the anions and the cations must be related by Eq. (14). Thus, the movement of all the ions may be described by the movement of either the cations or the anions. This fact simplifies matter considerably, because it allows us to consider the cations and the anions together as a single phase so that the tissue can be regarded as a triphasic medium. For this purpose, it is convenient to define a "neutral salt" [27] whose mass density ρ^i is given by:

$$\rho^i\equiv\rho^++\rho^- -\rho^F,\quad \rho^F=\tilde{c}^FM^+. \tag{33}$$

In Eq. (33), ρ^F is the mass density of that portion of Na^+ whose charges are exactly the same as the *FCD* c^F. Note that ρ^i is the mass per unit tissue volume, of Cl^- and an equal number of Na^+.

[6]Note in Eqs. (27), $\pi^+\equiv\pi^{+s}+\pi^{+w}+\pi^{+-}$, $\pi^-\equiv\pi^{-w}+\pi^{-+}+\pi^{-s}$, etc., the Lagrangian multiplier η_i appears in $\pi^{\alpha\beta}$ but not in π^α.

[7]This is related to the Onsager relations.

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From the zero current condition, Eq. (14), we obtain

$$\mathbf{v}^+ = \mathbf{v}^s + \frac{\tilde{c}^-}{\tilde{c}^+}(\mathbf{v}^- - \mathbf{v}^s) = \mathbf{v}^s + \frac{c^-}{c^+}(\mathbf{v}^- - \mathbf{v}^s). \tag{34}$$

Using Eq. (34), one can eliminate $\mathbf{v}^+$ from Eqs. (29)–(32) and obtain

$$-\rho^w \nabla \mu^w + K_{sw}(\mathbf{v}^s - \mathbf{v}^w) + K_{wi}(\mathbf{v}^- - \mathbf{v}^w) = \mathbf{0}, \tag{35}$$

$$-\rho^i \nabla \mu^i + K_{is}(\mathbf{v}^s - \mathbf{v}^-) + K_{wi}(\mathbf{v}^w - \mathbf{v}^-) = \mathbf{0}, \tag{36}$$

where

$$K_{ws} = f_{ws} + (f_{+w}/c^+)c^F,\ K_{wi}/c^- = (f_{w+}/c^+) + (f_{w-}/c^-),$$
$$K_{is} = (c^-/c^+)^2 f_{+s} + f_{-s} - (c^-/c^+)c^F(f_{w+}/c^+). \tag{37}$$

In arriving at Eqs. (35), we have assumed f_{+-} to be negligible. In Eq. (36), μ^i is the chemical potential for coions and anions defined by

$$M_i\mu^i = M^+\mu^+ + M^-\mu^- \tag{38}$$

where $M_i = M^+ + M^-$ is the molecular weight of NaCl and, as stated earlier, $\rho^i(\equiv \rho^+ + \rho^- - \rho^F)$ is the mass density of the neutral salt. Equations (35, 36) state that the chemical potentials are the sole driving forces for water and the neutral salt to overcome the frictional drag due to relative motions. That the chemical potential gradient is the driving force is well known in physicochemical theory [27].

Diffusion Equations for the Mobile Ions. From Eqs. (9), (10), and (33), one obtains that the neutral salt density $\rho^i = (M_i/M^-)\rho^-$, thus, the continuity equation for the neutral salt can be written:

$$\frac{\partial \rho^i}{\partial t} + \mathrm{div}(\rho^i \mathbf{v}^-) = 0. \tag{39}$$

In the absence of current, from Eq. (39) and the momentum equation (36), a diffusion equation for the neutral salt can be obtained as follows:

$$\mathrm{div}\left(\frac{\rho^{i2}}{K_{si}+K_{wi}}\nabla\mu^i\right) - \mathrm{div}\left(\rho^i\frac{K_{si}\mathbf{v}^s + K_{wi}\mathbf{v}^w}{K_{si}+K_{wi}}\right) = \frac{\partial \rho^i}{\partial t}. \tag{40}$$

This equation is needed for transient problems.

For isothermal conditions, the following Gibbs-Duhem relation can be derived:

$$-\frac{1}{2}\rho^s[\mathbf{K}^s tr(\mathbf{C}^{-1}\dot{\mathbf{C}}) - tr(\mathbf{F}^{-1}\mathbf{K}^s\mathbf{F}^{-T}\dot{\mathbf{C}})\mathbf{I}] + \rho^s\dot{\mathbf{K}}^s$$
$$+\rho^w\dot{\mu}^w\mathbf{I} + \rho^+\dot{\mu}^+\mathbf{I} + \rho^-\dot{\mu}^-\mathbf{I} + \dot{\boldsymbol{\sigma}} = 0, \tag{41}$$

where the total stress $\boldsymbol{\sigma}$ is defined as:

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^s + \boldsymbol{\sigma}^w + \boldsymbol{\sigma}^+ + \boldsymbol{\sigma}^-, \tag{42}$$

and "$\dot{}$" denotes a differential increment. In the absence of a solid phase, Eq. (42) reduces to the well known Gibbs-Duhem equation for a binary liquid solution:

$$\rho^w d\mu^w + \rho^i d\mu^i - dp = 0. \tag{43}$$

Summing the momentum Eqs. (15) over the four components, and making use of electroneutrality condition, Eq. (9), we obtain for equation of equilibrium for the entire mixture:

$$\mathrm{div}\boldsymbol{\sigma} = \mathbf{0}. \tag{44}$$

The Boundary of a Triphasic Mixture. It is natural to define the boundary of our triphasic mixture to be fixed to the solid matrix. This boundary, when it interfaces with an external bathing solution, either directly or through an idealized porous wall, allows free exchange of the interstitial fluid and ions with the external solution. For quasi-static conditions, the chemical potentials μ^i and μ^w are continuous across this boundary (see Appendix A for mathematical proof of this statement) and the total stress vector $(\boldsymbol{\sigma}\cdot\mathbf{n})$ of the mixture at such a boundary must equal the traction vector acting on the mixture as a result of balance of momentum. Thus, for the triphasic theory, it is natural to express the momentum equations in terms of the total stress $\boldsymbol{\sigma}$, Eq. (44), and of the two chemical potentials μ^w and μ^i, Eqs. (35) and (36).

Reference Configurations. In order to describe deformation, we need to specify a reference configuration. Although any configuration may serve as a reference, two are of particular interest: 1) unloaded equilibrium configuration in the hypothetical limit of infinite salt concentration; and 2) unloaded equilibrium configuration in a physiological saline bath. The physiological saline reference is often used for experimental measurements of cartilage swelling and deformation. However, for constitutive modeling of tissue swelling behavior, the reference corresponding to the infinite salt concentration is preferred, because at this reference the osmotic pressure and the chemical expansion pressure are negligibly small and the solid matrix may be considered to be stress free. In the following, all deformations are measured relative to this unloaded hypertonic state if not specifically stated.[8] This reference configuration was also used by Eisenberg and Grodzinsky [7, 8] to define the reference dimension of the tissue.

Specific Modeling Assumptions for Articular Cartilage. For cartilage, deformations due to ion induced swelling are generally small [1, 5, 6]. Thus, for the present theory, we shall use the infinitesimal strain tensor $\mathbf{E}$. Further, we assume the solid matrix to be isotropic,[9] and there is either no current or the effect of current is negligible. In this present development, we shall allow for large variations of ionic environment—from hypertonic condition to de-ionized water state—while the fixed charge groups per unit mass of the GAG chains are assumed not to change with the ionic environment. Note however, the *FCD* does change with deformation. For these assumptions, each of the Helmholtz energy functions (which are now functions of first and second invariants of $\mathbf{E}$, along with ρ^+, ρ^-, ρ^w and c^F) can be expanded into a polynomial:

$$A^\alpha \simeq f_1^\alpha + f_2^\alpha tr\mathbf{E} + f_3^\alpha(tr\mathbf{E})^2 + f_4^\alpha tr(\mathbf{EE}), \tag{45}$$

where f^α's are functions of ρ^w, ρ^+, ρ^-, c^F, and $\mathbf{E} = (\mathbf{C}-\mathbf{I})/2$. By substituting this expansion into Eqs. (26) and (42), we find, to the order of $0(\mathbf{E})$, the total mixture stress, the fluid stress (sum of water stress and the ion stresses) and the chemical potentials for the interstitial water and salt to be given by:

$$\boldsymbol{\sigma} = -p\mathbf{I} - T_c\mathbf{I} + \lambda_s tr(\mathbf{E})\mathbf{I} + 2\mu_s\mathbf{E}, \tag{46}$$

$$\boldsymbol{\sigma}^f \equiv \boldsymbol{\sigma}^w + \boldsymbol{\sigma}^+ + \boldsymbol{\sigma}^- = -\phi^f p\mathbf{I}, \tag{47}$$

$$\mu^w = \frac{1}{\rho_T^w}p + R_w + \frac{B_w}{\rho_T^w}tr(\mathbf{E}), \tag{48}$$

$$\mu^i = R_i + B_i tr(\mathbf{E}), \tag{49}$$

where $\phi^f \simeq \phi^w$ and all the coefficients T_c, λ_s, μ_s, R_w, R_i, B_w and B_i depend on ρ^α and c^F.[10] Equation (47) defines p as the fluid pressure on the fluid phase (water plus ions). Constitutively, p is indeterminate due to the assumption of incompressiblity and it must be determined from a boundary value problem. For a tissue in equilibrium under free swelling without external load, this fluid pressure is the osmotic pressure. From Eq. (46), we see that the mixture formulations lead to the decomposition of the total stress on the tissue into three contributions: the pressure p, the elastic stress $[\lambda_s tr(\mathbf{E})\mathbf{I} + 2\mu_s\mathbf{E})]$, and $-T_c\mathbf{I}$. We note that while p exists on both fluid and solid

[8]Change of reference configuration can be easily made. We shall do so later to relate our experimental measurements made in the physiological saline reference state.

[9]Up to this point, our theory is not limited to the infinitesimal strain theory or homogeneous, isotropic materials.

[10]The biphasic model is recovered when $T_c = R_w = B_w = R_i = B_i = 0$ in Eqs. (46)–(49). Note in this case, the chemical potential of water is simply p/ρ_T^w.

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phases (as it should, since both phases are incompressible), T_c exists only on the solid phase and it depends explicitly on c^F. We call T_c the chemical-expansion stress, which is due to the presence of the charge-to-charge electrostatic repulsive forces exerting on the PG-collagen network (the solid phase). It is important to note that for a tissue under load, the fluid pressure p not only depends on the Donnan osmotic effect, it also depends on loading rate as well as the load.

Equations (48) and (49) give the general forms of the chemical potentials for the water and ion phases. Determination of the specific forms for R_w, R_i, B_w, and B_i in terms of physicochemical parameters are given in the next three sections.

Chemical Potentials for the External NaCl Solution. The external solution is a mixture of two species, NaCl and water. With $\nu^*_{Na^+}$ and $\nu^*_{Cl^-}$ denoting the molar chemical potentials for Na^+ and Cl^-, respectively, and neglecting the small specific volumes of the ions, we have [27]:

$$\nu^*_{Na^+} = \nu^{o*}_{Na^+} + RT\ln(a^*_{Na^+}), \tag{50}$$

$$\nu^*_{Cl^-} = \nu^{o*}_{Cl^-} + RT\ln(a^*_{Cl^-}), \tag{51}$$

where $a^*_{Na^+}$ and $a^*_{Cl^-}$ are the activities of Na^+ and Cl^- which are related to their molar fractions x^*'s and activity coefficients γ^*'s by:

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$$a^*_{Na^+} = (\gamma^*_{Na^+})x^*_{Na^+}, \text{ and } a^*_{Cl^-} = (\gamma^*_{Cl^-})x^*_{Cl^-}, \tag{52}$$

where

$$x^*_{Na^+} = \frac{c^*_{Na^+}}{c^*_{Na^+} + c^*_{Cl^-} + c^*_w} \simeq \frac{c^*_{Na^+}}{c^*_w}, \text{ and } x^*_{Cl^-} \simeq \frac{c^*_{Cl^-}}{c^*_w}.$$

Here $c^*_{Na^+}$, $c^*_{Cl^-}$, and c^*_w are the molar concentrations of the Na^+ and Cl^- ions and water, respectively. The chemical potential per unit mass of NaCl, denoted by μ^{i*}, is then given by:

$$\mu^{i*} = \frac{\nu^*_{Na^+} + \nu^*_{Cl^-}}{M_i} = \frac{\nu^{o*}_{Na^+} + \nu^{o*}_{Cl^-}}{M_i} + \frac{RT}{M_i}\ln[(a^*_{Na^+})(a^*_{Cl^-})]$$
$$= \mu^{i*}_o + \frac{RT}{M_i}\ln[\gamma^{*2}_{\pm}c^{*2}], \tag{53}$$

where M_i is the molecular weight of NaCl, c^* ($= c^*_{Na^+} = c^*_{Cl^-}$) is the molar concentration of NaCl. We note that c^*_w, being essentially a constant ($\simeq 55.6$), is absorbed in μ^{i*}_o in the last expression in Eq. (53). Also in Eq. (53), $\gamma^{*2}_{\pm}$ is known as the mean activity coefficient of NaCl, whose value is well known and tabulated in [34]. At the physiological concentration of $c^* = 0.15\ M$, $\gamma^*_{\pm} \simeq 0.755$.

By integrating the Gibbs-Duhem equation for a binary solution, Eq. (43), we obtain the chemical potential for water as

$$\mu^{w*} = \mu^{w*}_o + \frac{p^*}{\rho^w_T} - \frac{RT}{\rho^w_T}2c^* - \frac{RT}{\rho^w_T}\int^{c^*} 2c^* d(\ln\gamma^*_{\pm})$$
$$\equiv \mu^{w*}_o + \frac{p^*}{\rho^w_T} - \frac{RT}{\rho^w_T}2\phi^* c^*, \tag{54}$$

where ϕ^* is the osmotic coefficient and it is related to the mean activity coefficient $\gamma^*_{\pm}$ as shown. The value of ϕ^* is also well known and tabulated; it decreases monotonically from 1.0 at $c^* = 0$ to 0.92 at $c^* \simeq 1\ M$ [34].

Chemical Potentials for a Charged Macromolecular Solution. For uni-uni salt, with c denoting the concentration of the neutral salt, we have $c_{Na^+} = c + c^F$ and $c_{Cl^-} = c$, so that for the neutral salt,[11]

$$\mu^i = \mu^i_o + \frac{RT}{M_i}\ln[\gamma^2_{\pm}c(c + c^F)], \tag{55}$$

and for water

$$\mu^w = \mu^w_o + \frac{1}{\rho^w_T}[p - RT\phi(2c + c^F)], \tag{56}$$

where $\mu^i_o = \mu^{i*}_o$. We note that both the charged macromolecular solution (e.g., proteoglycan solution) and the simple binary solution of NaCl are the limiting cases of a triphasic medium with $B_i = B_w = 0$ in Eqs. (48) and (49). Thus,

$$R_i = \mu^i_o + \frac{RT}{M_i}\ln[\gamma^2_{\pm}c(c + c^F)], \tag{57}$$

and

$$R_w = \mu^w_o - \frac{RT}{\rho^w_T}\phi(2c + c^F). \tag{58}$$

Chemical Potentials for the Triphasic Medium. For cartilage, we assume that the activities of Na^+ and Cl^- are given by the same expression as those in the right side of Eq. (52), i.e.,

$$a_{Na^+} = (\gamma_{Na^+})x_{Na^+} \text{ and } a_{Cl^-} = (\gamma_{Cl^-})x_{Cl^-}, \tag{59}$$

where x_{Na^+} and x_{Cl^-} are the molar fractions. Since the total number of moles of all the components is essentially the number of mole of water, these molar fractions can be approximated by the ratios of the respective number of mole to the number of mole of water in the tissue, i.e., $x_{Na^+} = (M_w/\rho^w_T)c_{Na^+}$ and $x_{Cl^-} = (M_w/\rho^w_T)c_{Cl^-}$, where c_{Na^+} and c_{Cl^-} are concentrations per unit interstitial water in the tissue. Thus,

$$\mu^i = \tilde{\mu}^i_o + \frac{RT}{M_i}\ln[(a^*_{Na^+})(a^*_{Cl^-})] = \mu^i_o + \frac{RT}{M_i}\ln[\gamma^2_{\pm}c(c + c^F)], \tag{60}$$

where c denotes the moles of the neutral salt NaCl (or, of the anions Cl^-) per unit interstitial water volume, and $\mu^i_o = \mu^{i*}_o$. The activity coefficient $\gamma_{\pm}$ is in general a function of c and c^F.[12] Now by comparing this form of μ^i, Eq. (60), with the general form derived from continuum theory, Eq. (49), we obtain the coefficient $B_i = 0$ and R_i is given by the right side of Eq. (60). For the determination of R_w, we note that in the absence of a solid phase, Eqs. (48) and (49) must reduce to the chemical potentials for the macromolecular solution where R_i and R_w are given by Eq. (57) and Eq. (58). Therefore for the triphasic medium under infinitesimal deformation, the chemical potential for the interstitial water is:

$$\mu^w = \mu^w_o + \frac{1}{\rho^w_T}[p - RT\phi(2c + c^F) + B_w tr\mathbf{E}]. \tag{61}$$

The Chemical-Expansion Stress T_c. As already explained earlier, the chemical-expansion stress $-T_c(c, c^F)$ in Eq. (46) is a continuum representation of the stress exerted on the solid phase by actions other than the osmotic pressure and the elastic stresses. Equilibrium cartilage swelling is due to the combined action of this chemical-expansion stress and the Donnan osmotic pressure. A specific constitutive assumption is required to define this chemical-expansion stress. Below, we shall show how the free-swelling measurement [5, 6] provides a way to determine this stress.

Summary of Equations for the Triphasic Theory

In summary, the following constitutive equations are for an isotropic triphasic model in a single uni-uni salt bath (e.g., NaCl) where the fixed charge groups in the tissue remain con-

[11]The general expressions of the chemical potentials for a macromolecular solution with a salt $A_{\nu_+}B_{\nu_-}$ are

$$\mu^i = \mu^i_o + (RT/M_i)\ln[\gamma^{\nu_- + \nu_+}_{\pm}(\nu_- c)^{\nu_-}(c + FCD/z_+)^{\nu_+}],$$

and

$$\mu^w = \mu^w_o + [p - RT\phi(\nu_- c + \nu_+(c + FCD/z_+))]/\rho^w_T,$$

where z_+ is the valency of cations of the salt.

[12]Strictly speaking, $\gamma_{\pm}$ also depends on the solid dilatation, but this dependence is expected to be negligible.

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stant, the counter ions are the cations of the salt (e.g., Na^+), and that the deformation of the solid matrix is small:

(A) For the external salt solution, the chemical potentials are

$$\mu^{i*}=\mu_o^{i*}+\frac{RT}{M_i}\ln(\gamma_\pm^{*2}c^{*2}),\quad \mu^{w*}=\mu_o^{w*}+\frac{1}{\rho_T^w}(p^*-2RT\phi^*c^*). \tag{62}$$

(B) For articular cartilage, the constitutive equations and chemical potentials are

$$\sigma^f=\sigma^w+\sigma^++\sigma^-=-\phi^f p\mathbf{I} \tag{63}$$

$$\sigma=\sigma^f+\sigma^s=-(p+T_c)\mathbf{I}+\lambda_s(tr\mathbf{E})\mathbf{I}+2\mu_s\mathbf{E}, \tag{64}$$

$$\mu^i=\mu_o^i+\frac{RT}{M_i}\ln[\gamma_\pm^2c(c+c^F)],\quad \mu^w=\mu_o^w+\frac{1}{\rho_T^w}[p-RT\phi(2c+c^F)+B_w tr\mathbf{E}]. \tag{65}$$

We note that Eq. (64) is identical in form to that proposed by Myers et al. [5], and that by Eisenberg and Grodzinsky [7, 8].[13] However, interpretation of the pressure terms are different. In [5], p is to be interpreted as the pressure above the osmotic pressure, and in [7, 8] P_f is defined as the fluid pressure given by $P_f=P-\chi\Delta\pi$, where P is the hydrostatic pressure, χ is the Staverman reflection coefficient and $\Delta\pi$ is the osmotic pressure difference. The fluid pressure P_f defined in [7, 8] is different from the fluid pressure p in this paper (e.g., the equilibrium fluid pressure p in Eq. (69) below is osmotic pressure which is not the same as P_f of [7, 8]). In addition, references [7, 8] do not provide any means to calculate $\Delta\pi$. In our view, the meaning of the "fluid pressure" P_f as defined in [7, 8], and the boundary conditions on P_f as used in [7, 8] are difficult to comprehend and to justify.

(C) For the fixed charge density, c^F, the relationship to deformation is

$$c^F=c_o^F\left(1-\frac{1}{\phi_o^w}tr(\mathbf{E})\right), \tag{66}$$

describing its variation with solid matrix compression. Here ϕ_o^w is volume fraction of interstitial water, assumed to be homogeneous, and c_o^F is moles of fixed charges per unit interstitial water volume at the reference configuration which is assumed not to change with the ionic environment.

(D) In the absence of electric current, the momentum equation for the total triphasic mixture, the interstitial water and the neutral salt (or, of the anions Cl^-) are given by Eqs. (44), (35), and (36), respectively.

The above equations are valid for both equilibrium and nonequilibrium problems. For equilibrium problems, the momentum equations for the interstitial water and neutral salt, Eqs. (35) and (36) yield $\mu^w=$ constant and $\mu^i=$ constant throughout the tissue, and the momentum equation for the total tissue, Eq. (44), yields $\sigma=$ **constant** throughout the tissue. Their values are to be determined by the boundary condition in a specific equilibrium problem.

Application of the Infinitesimal Strain Triphasic Theory for Cartilage Swelling

Equilibrium Free-Swelling—Change of Dimension. Consider a specimen of tissue placed snuggly inside a circular confining ring (Fig. 2). No load is applied onto the specimen in the z-direction. The specimen is equilibrated in a single salt

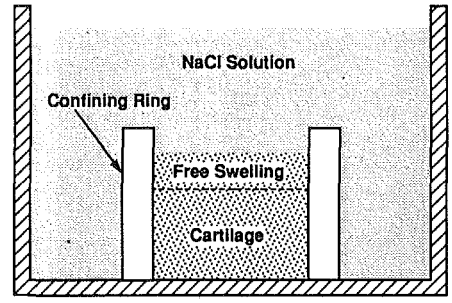

Fig. 2 Equilibrium free-swelling in the axial direction of a laterally confined cylindrical specimen of cartilage

(NaCl) solution with concentration c^*. Thus, the boundary conditions on the free surface are:

$$\sigma_{zz}=0,\ \mu^i=\mu^{i*},\ \text{and}\ \mu^w=\mu^{w*}. \tag{67}$$

These conditions, together with Eqs. (62), (64), and (65), and the fact that μ^i, μ^w, and σ_{zz} are constants throughout the tissue, yield the following three equations for the neutral salt concentration c, the fluid pressure p, and the swelling strain e_{zz} (relative to a hypertonic reference configuration):

$$c(c+c^F)=\left(\frac{\gamma_\pm^*}{\gamma_\pm}\right)^2c^{*2}, \tag{68}$$

$$p=RT[\phi(2c+c^F)-2\phi^*c^*]+P_\infty-B_we_{zz}, \tag{69}$$

$$(\lambda_s+2\mu_s)e_{zz}=p+T_c, \tag{70}$$

where c^F is related to e_{zz} through Eq. (66). In Eq. (69), $P_\infty=-\rho_T^w(\mu_o^w-\mu_o^{w*})$. We see from Eq. (68), our triphasic theory yields the Donnan equilibrium ion distribution equation. Equation (68) (in molality form) has been used extensively to determine the physico-chemical properties of articular cartilage [2–4]. From Eq. (69), we see three sources contribute to the total fluid pressure at equilibrium: 1) Donnan osmotic pressure $RT[\phi(2c+c^F)-2\phi^*c^*]$; 2) osmotic pressure due to the presence of the proteoglycans, P_∞; and 3) solid matrix compression $-B_we_{zz}$, where B_w is a coupling coefficient. The latter two sources embody the so-called "excluded volume effect" [12]. In the absence of the solid matrix, i.e., $B_w=0$, Eq. (69) yields the same osmotic pressure for a proteoglycan solution in equilibrium with a NaCl solution [4, 12, 13]. The term P_∞, arises from a difference of the reference chemical potentials of water between the internal (μ_o^w) solution and the external (μ_o^{w*}) solution which depends on the concentration of the proteoglycans in the tissue. By Taylor series expansion, P_∞ can be expressed as:

$$P_\infty=\phi_{pg}RT(c_{pg}/M_{pg})+0[(c_{pg}/M_{pg})^2], \tag{71}$$

where c_{pg} is the proteoglycan concentration (in g/ml), M_{pg} is the mean molecular weight and ϕ_{pg} is the osmotic coefficient. This expression may be found in [28] for osmotic pressure of macromolecular solutions.

Osmotic pressures of PG solutions, as a function of FCD, have been measured by Urban and co-workers [12, 13]. The measurements were made through an equilibration of the (PG, NaCl) solution (internal) with a (polyethlene glycol, NaCl) solution (external) of known osmotic pressure. These investigators found, for example at $FCD=0.4$ mEq/(g water), $p=0.25$ MPa at 0.15M NaCl, and $p=0.12$ MPa at 1.5M NaCl. They also found that the Donnan ionic contribution to the osmotic pressure is over 85 percent of the total measured osmotic pressure.

For articular cartilage, a solid matrix exists. The description of its swelling behavior (changes in dimensions, volume and pressures) thus must also incorporate the deformation of the

[13] In references [7,8], $\sigma=-[P_f+\beta(c)]\mathbf{I}+\lambda(tr\mathbf{E})\mathbf{I}+2G\mathbf{E}$.

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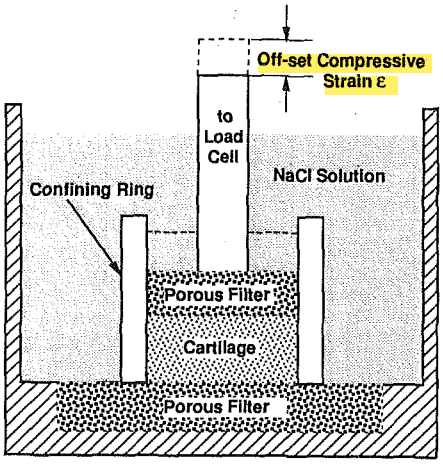

Fig. 3 Equilibrium confined swelling of a confined cylindrical specimen of cartilage. The sample is kept at a fixed volume corresponding to a prescribed off-set compressive strain ϵ in the axial direction.

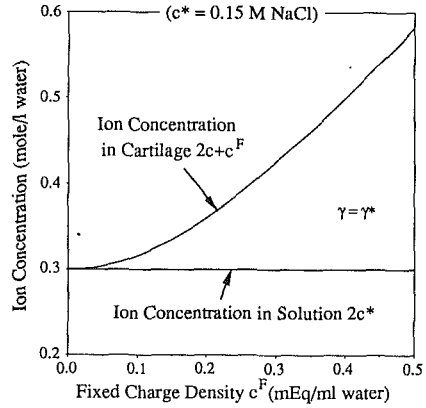

Fig. 4 Ion concentration in the interstitium ($2c+c^F$) versus *FCD* (c^F), *with the activity coefficients* $\gamma_\pm/\gamma_\pm^* = 1$. *The external bathing solution* NaCl concentration $c^* = 0.15$ *M*.

solid matrix. Equation (70) from this free-swelling problem states that a balance must exist between the fluid pressure p (including Donnan osmotic effect, P_∞, and the dilatation effect $B_w e_{zz}$), the chemical-expansion pressure T_c, and the elastic stress in solid matrix due to deformation ($(\lambda_s+2\mu_s)e_{zz}$). Equation (70) suggests that we define a total swelling pressure P_s to be the sum of the fluid pressure p and the chemical-expansion pressure T_c:

$$P_s = p + T_c. \tag{72}$$

Again, this definition for the swelling pressure is similar to that proposed by Maroudas [3]. According to Maroudas [3]: "$P_{\text{swelling}} = P_{\text{Donnan}} + P'_{\text{swelling}}$ where P'_{swelling} = swelling pressure due to effects other than ionic." However, from the present theory, the "effects other than ionic" contributing to the swelling pressure of cartilage are now explicitly defined in terms of T_c, $B_w tr\mathbf{E}$ and P_∞. Also the present theory is a general 3-dimensional tensorial theory capable of predicting all the deformational and flow fields.

From Eqs. (66) and (68), we see that under equilibrium c and c^F are functions of c^* and e_{zz}, therefore p and T_c are functions of c^* and e_{zz} also. Thus, for a given concentration c^* of the bathing solution, Eq. (70) can be linearized in terms of the infinitesimal strain e_{zz}, using a Taylor series expansion. This yields an explicit relationship between the free-swelling strain e_{zz} and c^* given by:

$$H_A e_{zz} = \{RT[\phi(2c+c^F) - 2\phi^* c^*] + P_\infty + T_c\}|_{e_{zz}=0}, \tag{73}$$

where

$$H_A = \lambda_s + 2\mu_s - \frac{d}{de_{zz}}[p+T_c]|_{e_{zz}=0}. \tag{74}$$

Equation (73) states that the elastic stress in the solid matrix, i.e., $H_A e_{zz}$, from free-swelling depends on the external salt concentration c^* and the fixed charge density c^F.

We will show in a section below that the coefficient H_A, as a function of c^*, is the aggregate modulus used in the linear biphasic theory for articular cartilage [14–16] determined in an external bath concentration c^*. The biphasic aggregate modulus has been measured by various authors [6, 7, 14, 35, 36]. Further, with H_A and the total swelling pressure P_s at c^* determined at $e_{zz}=0$ from confined compression experiments, Eqs. (73) and (74) can be used to calculate the chemical-expansion stress $-T_c$ from measured values of e_{zz} under free swelling conditions. An example of this will be presented below.

Equilibrium Confined-Swelling—Change of Stress. In the confined isometric swelling experiment with the specimen equilibrated in c^*, the volume of the cylindrical sample is prevented from changing by prescribing the fixed axial compressive offset strain $\epsilon(=-e_{zz})$, Fig. 3. The platen compressing the specimen is permeable to the ion and fluid phases. At equilibrium, the momentum equations and boundary conditions again yield the Donnan equilibrium ion distribution, i.e., Eq. (68), and again the fluid pressure:

$$p = RT[\phi(2c+c^F) - 2\phi^* c^*] + P_\infty + B_w\epsilon. \tag{75}$$

But now, from the boundary condition $\sigma_{zz} = -P_{app}$ and the constitutive relation Eq. (64), we obtain:

$$P_{app} = (\lambda_s + 2\mu_s)\epsilon + P_s. \tag{76}$$

This equation states that the total applied compressive stress required to keep the tissue at the isometric configuration is the sum of the elastic compressive stress $(\lambda_s+2\mu_s)\epsilon$ and the total swelling pressure, $P_s = p + T_c$. This equilibrium relation is similar in form to that developed in Eisenberg and Grodzinsky [7, 8], though the interpretation of the fluid pressure term is different.

Some Examples for Cartilage Swelling Using Infinitesimal Strain Theory

Donnan Ion Distribution. Equation (68) defines the equilibrium neutral salt NaCl concentration c in the tissue as a function of c^F and c^*. The ion concentration (Na^+ and Cl^-) inside the cartilage is $2c+c^F$, and the ion concentration in the external bath is $2c^*$. The difference between these two concentrations may be calculated for given values of c_o^F, ϕ_o^w and ϵ_{offset}. Figure 4 gives the values of the internal ion concentration $(2c+c^F)$ for $c^* = 0.15$ *M* (with total concentration of the ions in the bathing solution to be 0.30 *M*), and for values of c_o^F ranging from 0 to 0.5 mEq/(ml water) and the ratio $\gamma_\pm^*/\gamma_\pm$ taken to be unity [2, 3]. Further, the tissue is kept at the reference configuration ($\epsilon_{\text{offset}}=0$). The curve shows clearly that the ion concentration $2c+c^F$ in the tissue is always larger than the concentration $2c^*$ in the bath and it increases nonlinearly as c^F increases. This imbalance of ion concentrations inside and outside the tissue gives rise to the Donnan ionic osmotic pressure.

Donnan Osmotic Pressure Versus Total Swelling Pressure. Figure 5 shows the variation of Donnan osmotic pressure for a tissue equilibrated in a 0.15 *M* NaCl solution as a function of c^F. The tissue volume is kept fixed at its reference configuration $e_{zz}=0$ (i.e., the volume of the unloaded tissue

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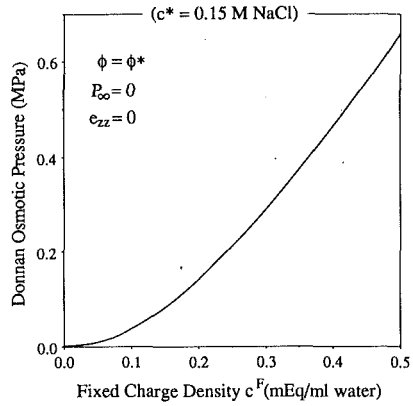

Fig. 5 Donnan osmotic pressure (p) versus *FCD* (c^F) for an external solution of 0.15 *M* NaCl. The specimen is kept at its hypertonic reference configuration ($e_{zz}=0$). The activity and osmotic coefficients are assumed to satisfy $\gamma_\pm/\gamma_\pm^*=1$ and $\phi/\phi^*=1$, with the values of ϕ^* taken from [34], and the colligative contribution of PG is neglected, i.e., $P_\infty=0$.

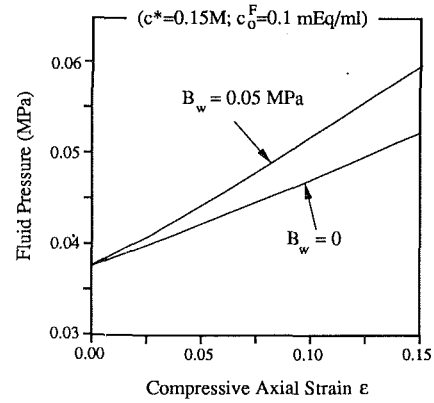

Fig. 6 Fluid pressure p versus compressive strain ϵ for two values of B_w: 0 and 0.05 MPa. The *FCD* and the fluid content in cartilage at its hypertonic reference configuration ($e_{zz}=0$) are assumed to be 0.1 mEq/ml and 80 percent, respectively.

in the hypertonic solution). For these calculations, the osmotic coefficient ϕ is assumed to be the same as ϕ^* for the NaCl bath. The values of ϕ^* were taken from available data [34], and P_∞ was assumed to be negligible. We see that the Donnan (ionic) osmotic pressure increases nonlinearly with c^F to approximately 0.6 Mpa at $c^F=0.5$ (mEq/ml water). For a coupling coefficient $B_w=0.05$ MPa, $c^*=0.15$ *M* NaCl, and $c^F=0.10$ (mEq/ml water), Fig. 6 shows the effect of the solid matrix compressive strain on fluid pressure, Eq. (75), where the strain is measured from the hypertonic reference configuration.

The total applied compressive stress P_{app} needed to keep the tissue at the compressive strain ϵ is given by Eq. (76). This applied compressive stress is balanced by two stress components: 1) the elastic compressive stress $(\lambda_s+2\mu_s)\epsilon$; and the total swelling pressure $P_s=p+T_c$. To compare our equation with that of Maroudas [3,4], we rewrite Eq. (76) as:

$$P_{app}=P_s-T_e, \tag{77}$$

where $T_e=-(\lambda_s+2\mu_s)\epsilon$ is the elastic tensile stress. This equation is similar to that provided by Maroudas, who called P_{app} the "applied hydrostatic pressure" in reference [3] or "net swelling pressure" in reference [4], P_s the "swelling pressure" in reference [3] or "proteoglycan osmotic pressure" in reference [4], and T_e the "hydrostatic pressure produced by elastic stress" in reference [3] or "collagen network tensile stress" in reference [4].

From the triphasic theory, we see that when the offset compression strain $\epsilon=0$, then $P_{app}=P_s$. This means that for a tissue bathed in an external salt solution of concentration c^*, the applied stress needed to keep the tissue at the unloaded hypertonic reference configuration dimensions is completely balanced by the total swelling pressure; and there is no elastic stress. This swelling pressure has been reported as 0.17 MPa for a plug of cartilage from a human femoral head [2, 3]. However, it should be noted again that in general the total swelling pressure is the sum of two components: 1) the fluid pressure (which includes the Donnan osmotic pressure, the solid matrix coupling term $B_w\epsilon$ and P_∞); and 2) the chemical-expansion stress $-T_c$.

Dependence of the Aggregate Modulus H_A on c^* and c^F. In the triphasic stress-strain relationship for the solid matrix, Eq. (64), the intrinsic material coefficients λ_s, μ_s, and B_w depends on c^* and c^F. An explicit relation may be derived, Eq. (81) below, equating the biphasic aggregate modulus used in [14–16] to H_A defined in Eq. (74) at c^*. To do this, we first obtain

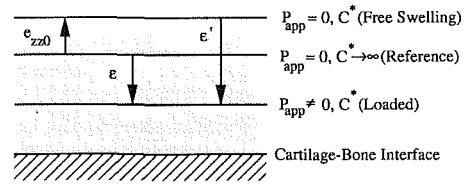

Fig. 7 Reference states from which ϵ and ϵ' are measured

a linear relationship between the applied stress P_{app} and the applied strain compressive ϵ from Eq. (76) by using Taylor series expansions of P_s, Eqs. (72) and (75), in terms of e_{zz}, we obtain:

$$P_{app}=H_A\epsilon+\{RT[\phi(2c+c^F)-2\phi^*c^*]+T_c(c,c^F)+P_\infty\}|_{e_{zz}=0}, \tag{78}$$

where H_A is given in Eq. (74) and as indicated, the terms in { } are evaluated at $e_{zz}=0$. As shown in Fig. 7, the axial strain can be measured from two reference configurations: (i) ϵ, measured from the unloaded hypertonic state, or (ii) ϵ', measured from the unloaded but pre-stressed state in a given bathing solution of concentration c^*. The two strains ϵ and ϵ' are related by the simple relationship: (Fig. 7)

$$\epsilon=\epsilon'-e_{zzo}, \tag{79}$$

where e_{zzo} is the free swelling axial tensile strain (measured from the hypertonic reference configuration) for the unloaded tissue at c^* given by Eqs. (68)–(70):

$$e_{zzo}=H_A^{-1}\{RT[\phi(2c+c^F)-2\phi^*c^*]+T_c+P_\infty\}|_{e_{zz}=0}. \tag{80}$$

We note that under compressive loading P_{app} at c^*, ϵ' is always positive, whereas ϵ can be either negative ($P_{app}<P_s$) or positive ($P_{app}>P_s$), see Eq. (76). Substituting Eqs. (79) and (80) into (78), we obtain:

$$P_{app}=H_A\epsilon', \tag{81}$$

which is the equilibrium stress-strain relationship under the confined compression condition, and H_A is the aggregate modulus as used in the biphasic theory at c^*. With this result, using the experimental measurements of H_A and ϵ', we can now determine the chemical-expansion stress $-T_c$, for known physico-chemical values of P_∞, $\gamma_\pm$, ϕ, c_o^F and ϕ_o^w.

Chemical-Expansion Stress $-T_c$. Table 2 shows the data for H_A in a confined compression experiment and strain ϵ' in a one-dimensional free-swelling experiment of bovine articular cartilage for six NaCl concentrations c^* [6], where ϵ' is meas-

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Table 2 Experimental measurements [6] of biphasic aggregate modulus H_A' and free-swelling axial strain (measured from the unloaded configuration in distilled water) of confined cartilage specimen

c* (M NaCl)	0.0	0.015	0.15	0.30	0.75	1.50	2.50
Aggregate modulus H_A (MPa)	2.00	–	1.00	0.80	0.58	0.50	0.48
Axial free-swelling strain (measured from unloaded configuration in distilled water)	0.0	−0.121	−0.145	−0.208	−0.245	−0.250	−0.265

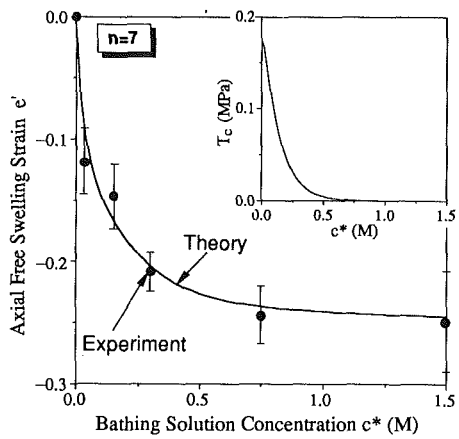

Fig. 8 Experimental ( ) [6] and curvefit swelling results, Eqs. (68)-(70), of a specimen of cartilage which is axially free and laterally confined (the point corresponding to $c^* = 0$ denotes distilled water). The chemical-expansion function T_c for this case is shown in the inserted figure, with parameters $a_o = 2.5$ (MPa·ml/mEq) and $\kappa = 7.5$ M^{-1} from curvefitting. Other parameters used are the same as used for Figs. 4-6.

ured from the unloaded configuration at $c^* = 0$ (water). Both H_A and ϵ' decrease with c^* in an exponential manner. Thus, it seems natural to assume T_c also to decrease exponentially with c^*. Also, since the chemical expansion effect must be absent when c^F is zero, therefore, the exponential term is multiplied by $a_o\, c^F$ where a_o is constant.[14] However, since c^* is related to c and c^F by Eq. (68), T_c can be written in terms of c and c^F as[15]

$$T_c(c,c^F) = a_o c^F e^{-\kappa(\gamma_\pm/\gamma_\pm^*)\sqrt{c(c+c^F)}} - P_\infty. \tag{82}$$

Inserting this form for T_c into Eq. (73), using the reference transformation Eq. (79) for strain (where e_{zzo} corresponds to $c^* = 0$) and assuming $c_o^F = 0.1$ mEq/(ml water), $\gamma_\pm/\gamma_\pm^* = 1$, $\phi/\phi^* = 1$, and with $\phi^*(c^*)$ given by [34], the values for a_o and κ may be obtained from curve-fitting the data given in Table 2.[16] The result from this calculation yields $a_o = 2.5$ MPa/(mEq/ml) and $\kappa = 7.5$ m^{-1}. With this chemical-expansion stress, the free-swelling strain may also be calculated and a comparison of this curvefit with the measured free-swelling strain [6] is shown in Fig. 8. Surprisingly the infinitesimal strain theory seems able to predict swelling behavior as high as 25%. We note that the point of $c = 0$ is to be regarded as corresponding to a very dilute salt solution in Figs. 8 and 9.

When the tissue is kept at the hypertonic reference configuration, $\epsilon = 0$, the fluid pressure p at any c^* is the Donnan (ionic) osmotic pressure, see Eq. (75) (with negligible P_∞). From our results for p and $-T_c$, we may now calculate their relative contributions in producing the total swelling pressure

[14]Higher order term in c^F can be added if it is found necessary for accuracy or for predicting new phenomena.

[15]The term P_∞ is included so that $e_{zz} = 0$ as $c_{\to\infty}$ as required by our choice of reference configuration.

[16]We assume that H_A for compression and tension are the same.

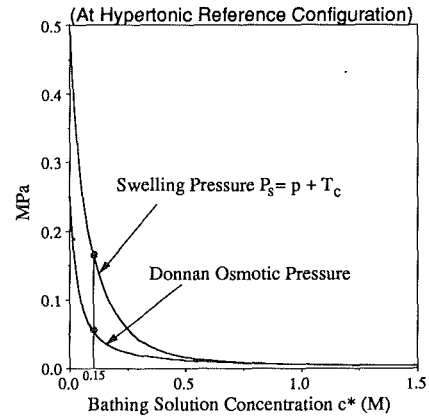

Fig. 9 Donnan osmotic pressure and total swelling pressure of cartilage (at the hypertonic reference configuration) as functions of the bathing solution NaCl concentration. Parameters used in calculating p and T_c are the same as used for Figs. 4-6 and 8.

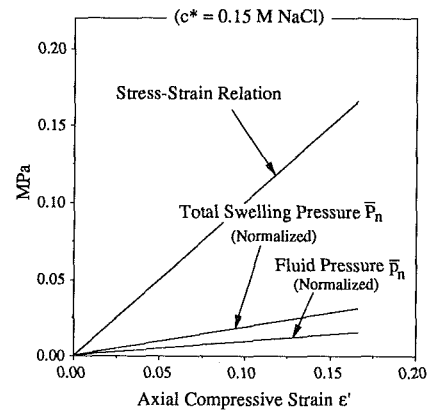

Fig. 10 The stress-strain curve for the tissue properties from [6] at $c^* = 0.15$ M NaCl bathing solution. Slope is the biphasic aggregate modulus H_A'. The axial compressive strain ϵ' is measured from the unloaded configuration. Also shown are the predicted and normalized total swelling pressure $\overline{P}_n = P_s - P_s|_{\epsilon'=0}$ and fluid pressure $\overline{p}_n = p - p|_{\epsilon'=0}$ as functions of the compressive strain.

P_s. This calculation is shown in Fig. 9. For our physico-chemical coefficients, at $c^* = 0.15$ M NaCl, we find that the Donnan (ionic) osmotic pressure and the chemical-expansion stress contribute in nearly equal proportions to the total swelling pressure P_s.

Contribution of Physico-chemical Factors to Cartilage Stiffness. The role of physico-chemical factors in determining cartilage compressive stiffness may also be calculated. Figure 10 shows a linear stress-strain curve for the confined compression problem when the specimen is equilibrated in a 0.15 M NaCl, using $H_A = 1$ MPa (Table 2), and the assumed physico-chemical coefficients. Again, in this figure, the compressive strain ϵ' is measured with reference to the unloaded configuration at $c^* = 15$ M. The solid curve is P_{app} versus ϵ'. Also the normalized total swelling pressure $\overline{P}_n = P_s - P_s|_{\epsilon'=0}$ and the normalized fluid pressure, $\overline{p}_n = p - p|_{\epsilon'=0}$ are presented. The slope of the stress-strain curve is the aggregate modulus H_A at c^* of the linear biphasic theory. From Eq. (74) we have

$$H_A = \lambda_s + 2\mu_s + \frac{dP_s}{d\epsilon'}\Big|_{\epsilon'=e_{zzo}}. \tag{83}$$

Thus, the total swelling pressure P_s contributes to the tissue stiffness H_A through its slope $dP_s/d\epsilon'$ at the hypertonic ref-

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erence configuration. Figure 10 shows P_s to be essentially linear in compressive strain ϵ' and its slope is relatively small in comparison with the slope of the stress-strain relationship. Thus for this set of bovine articular cartilage specimens, assuming $c_o^F = 0.1$ mEq/(ml water), the intrinsic stiffness of the solid matrix contributes a major part to the overall compressive stiffness of cartilage [1, 6, 7, 8, 35, 36].

Conclusion

A theory for a tertiary mixture has been developed to model articular cartilage swelling and deformational behaviors. This triphasic theory includes two fluid-solid phases (biphasic), and an ion phase with two ionic species (Na^+ and Cl^-). The fixed charge groups per unit mass of the solid phase are assumed to remain unchanged and the counter ions are the cations of a single salt in its solution the tissue is immersed. This theory can be applied to equilibrium as well as transient problems. It is capable of predicting the stress-strain fields in the solid matrix, interstitial fluid flow, along with the ion distribution and fluid pressure. For equilibrium problems, an expression for the total equilibrium swelling pressure P_s is derived, and this is the sum of two effects: 1) osmotic pressure; and 2) chemical-expansion stress. Osmotic pressure derives predominantly from the Donnan ionic effect due to an excess of total ion concentration within the tissue with fixed charged density c^F over the ion concentration in the external solution. The chemical-expansion stress T_c derives from the charge-to-charge repulsive force between the closely spaced charged groups along the glycosaminoglycan chains of the proteoglycans. Analytical terms for other factors, which may be interpreted as excluded-volume effects, have also been derived. The theory is capable of predicting all field variables within the tissue when it is subjected to either a mechanical load and/or a chemical load.

The momentum equations for the neutral salt and the interstitial water are expressed in terms of their chemical potentials μ^i and μ^w. In a NaCl bath, these chemical potentials are given by $\mu^i = \mu_o^i + (RT/M_i)\ln[\gamma_\pm^2 c(c+c^F)]$ and $\mu^w = \mu_o^w + [p - RT\phi(2c+c^F) + B_w e]/\rho_T^w$. For the isotropic model, the constitutive equation for the total stress tensor is $\mathbf{T} = -(p+T_c)\mathbf{I} + \lambda_s e\mathbf{I} + 2\mu_s\mathbf{E}$, which depends nonlinearly on the concentration c. Two preferred configurations corresponding to two external salt concentrations (hypertonic and physiological) have been defined. The calculated strain in one configuration can be easily transformed to the strains in the other configuration. The theory has been applied to describe the equilibrium free swelling and confined compression behavior. Some numerical calculations, based on a set of experimental data, are given to demonstrate how the theory can be used to assess the relative contributions of the Donnan (ionic) osmotic effect and the chemical-expansion effect to the total compressive stiffness of articular cartilage. The triphasic theory encompasses both the physico-chemical view of cartilage swelling, and the biphasic view of solid matrix deformation and interstitial fluid flow.

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APPENDIX

The interface boundary conditions for the total mixture stress and the fluid chemical potentials are derived in this appendix. Consider a boundary that is fixed to the solid matrix of a triphasic mixture, so that

$$[[\mathbf{v}^s]]\cdot\mathbf{n}=0, \tag{A1}$$

The jump conditions of our triphasic mixture corresponding to the conservations of mass, momentum, energy and entropy, with inertia neglected, on such a boundary are [37]

$$[[\rho^w(\mathbf{v}^w-\mathbf{v}^s)]]\cdot\mathbf{n}=0, \tag{A2}$$

$$[[\rho^\pm(\mathbf{v}^\pm-\mathbf{v}^s)]]\cdot\mathbf{n}=0, \tag{A3}$$

$$[[\sigma]]\cdot\mathbf{n}=\mathbf{0}, \tag{A4}$$

$$[[\mathbf{q}^s+\mathbf{q}^f+\mathbf{q}^\pm+\mathbf{v}^s\cdot\sigma^s+\mathbf{v}^w\cdot\sigma^w+\mathbf{v}^\pm\cdot\sigma^\pm+\rho^w e^w(\mathbf{v}^w-\mathbf{v}^s)+\rho^\pm e^\pm(\mathbf{v}^\pm-\mathbf{v}^s)]]\cdot\mathbf{n}=0, \tag{A5}$$

$$\left[\left[\frac{1}{T}(\mathbf{q}^s+\mathbf{q}^f+\mathbf{q}^i)+\rho^w\eta^w(\mathbf{v}^w-\mathbf{v}^s)+\rho^\pm\eta^\pm(\mathbf{v}^\pm-\mathbf{v}^s)\right]\right]\cdot\mathbf{n}=0. \tag{A6}$$

For isothermal process, one obtains, with the definition of chemical potentials, from Eqs. (A4)–(A6):

$$[[\mu^w\rho^w(\mathbf{v}^w-\mathbf{v}^s)+\mu^+\rho^+(\mathbf{v}^+-\mathbf{v}^s)+\mu^-\rho^-(\mathbf{v}^--\mathbf{v}^s)]]\cdot\mathbf{n}=0 \tag{A7}$$

In the absence of current, by Eq. (14), Eq. (A7) can be written as

$$[[\mu^w\rho^w(\mathbf{v}^w-\mathbf{v}^s)+\mu^i\rho^i(\mathbf{v}^i-\mathbf{v}^s)]]\cdot\mathbf{n}=0. \tag{A8}$$

Since the chemical potentials μ^w and μ^i are constitutively independent of the relative velocities $\mathbf{v}^w-\mathbf{v}^s$ and $\mathbf{v}^i-\mathbf{v}^s$; also $\mathbf{v}^w-\mathbf{v}^s$ and $\mathbf{v}^i-\mathbf{v}^s$, ρ^w and ρ^i are independent variables, so that μ^w is independent of $\rho^w(\mathbf{v}^w-\mathbf{v}^s)$ and μ^i is independent of $\rho^i(\mathbf{v}^i-\mathbf{v}^s)$. Thus, from Eqs. (A2) and (A3) one can conclude that

$$[[\mu^w]]=0, \text{ and } [[\mu^i]]=0. \tag{A9}$$

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